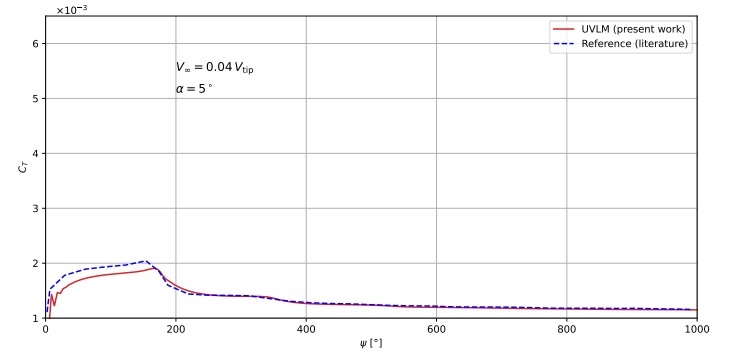
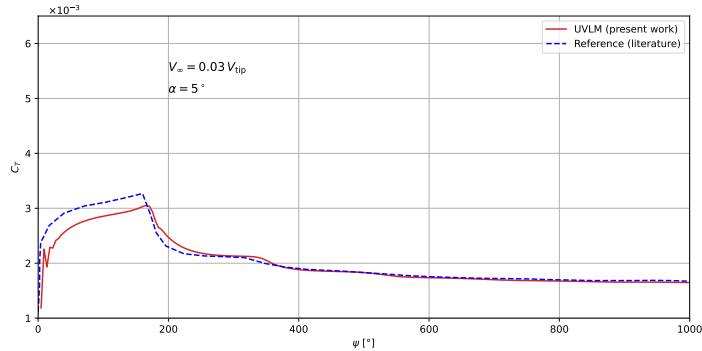
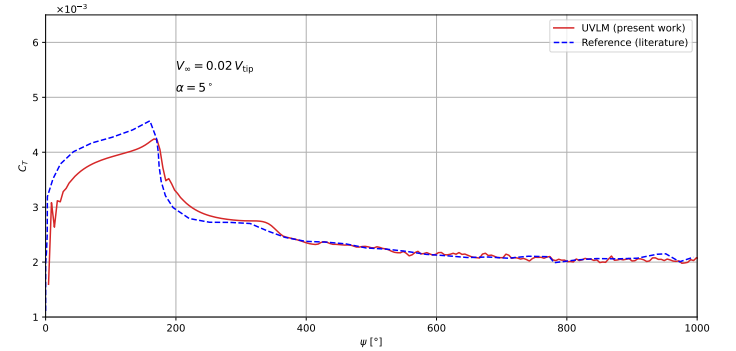
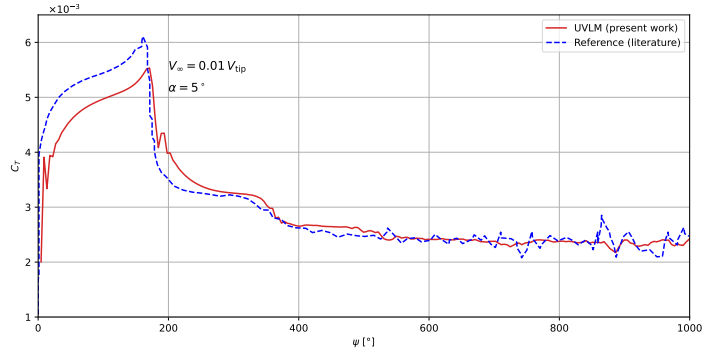


Effect of vertical freestream velocity on thrust coefficient for a two-blade rotor in hover ($\alpha = 5^\circ$)



The figures show the evolution of the thrust coefficient C_T as a function of the blade rotation angle ψ for an angle of attack of 5° and different vertical freestream velocities. The presence of a freestream velocity V_∞ along the rotation axis leads to more stable responses. This velocity convects the wake vortex segments away from the rotor, preventing wake overlap and causing the wakes to develop into spiral structures.

As a result, the noise in the response progressively decreases as the vertical velocity increases, eventually disappearing in the highest- V_∞ cases. The predicted behavior is consistent with that reported in the literature, with excellent agreement in the steady-state value. These results suggest that the discrepancies observed for $V_\infty = 0$ are primarily associated with wake-interaction numerical instabilities rather than with deficiencies of the aerodynamic model.